

1

INTRODUCTION

A battery manufacturer claims that the average lifetime of its batteries is 500 hours. To verify this claim, a hypothesis test is conducted using a sample of battery lifetimes. Additionally, an OC curve is plotted to understand the behavior of the test across a range of true means

DATA

Battery lifetimes for 30 randomly selected batteries are given, along with parameters mean and std dev being the manufacturer's claim

METHODOLOGY

Question 1

We input known parameters and given data. We then perform a hypothesis test on the given lifetimes to determine if the manufacturer's claim holds up. Since variance is known, we use z test

Formula	Explanation
$SE = \frac{s}{\sqrt{n}}$	• SE is standard error • s is sample standard deviation • n is the number of elements in the sample

One Sample Z Test Formula

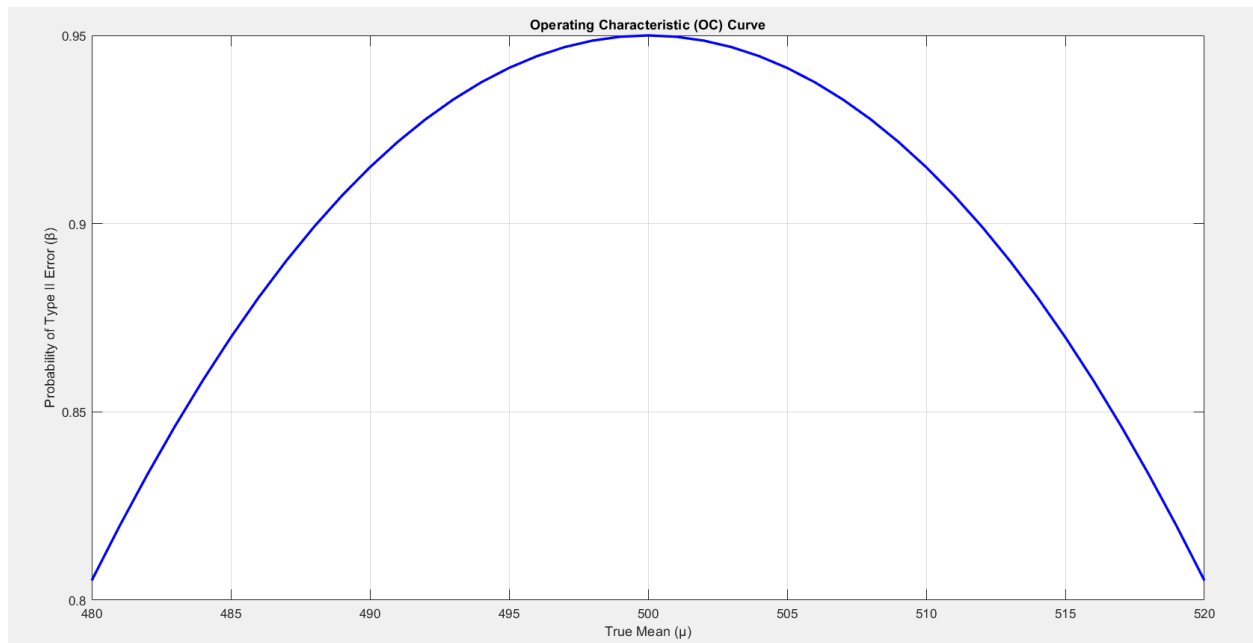
$$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

RESULTS

z: 0.1461

p-value: 0.8839

Fail to reject H0: No significant evidence of difference from 500 hours



DISCUSSION

The hypothesis test supports the manufacturer's claim, the sample data do not provide sufficient evidence to conclude that the average battery life is different from 500 hours (by observing the high p value and low z score) The OC curve gives additional insight into the performance of the test. It shows that the test is most likely to accept H_0 when the true mean is close to 500 and becomes less likely to do

so as the true mean moves further away. This confirms that the test has reasonable power near the extremes and low Type II error when the deviation is large. The shape of the OC curve is convex, which is expected for a two-sided z-test

CONCLUSION

The analysis indicates that there is no statistically significant difference between the claimed and observed mean battery life. The manufacturer's claim of a 500-hour average is consistent with the sample data. The OC curve further validates the reliability of the test, showing appropriate behavior across different true mean values

2

INTRODUCTION

Water usage is an important indicator in public health and resource management. A public health official claims that the mean daily home water use is 350 gallons. To assess the validity of this claim, a study was conducted using a sample of daily water usage from 20 randomly selected homes. Two statistical approaches are considered based on different assumptions about the population variance, where it is known, and unknown

DATA

20 random samples are given, along with parameters mean 350 gallons, $p = 0.05$, and population variance 144 (in known case)

METHODOLOGY

Question 2

We perform a two-tailed z-test followed by a two-tailed one-sample t-test with the following hypotheses

$H_0: \mu = 350$ (No difference in mean daily water use)

$H_1: \mu \neq 350$ (There is a difference)

$$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

$$t = \frac{\Sigma X - Y}{\frac{s}{\sqrt{n}}}$$

RESULTS

(a)

Sample mean: 353.80

Z statistic: 1.4162

P value: 0.1567

Fail to reject null hypothesis: Not enough evidence to contradict the claim

(b)

Sample mean: 353.80

Sample std dev: 21.85

T statistic: 0.7778

P value: 0.4462

DISCUSSION

Both tests yield consistent results, there is no statistically significant evidence that the mean water usage differs from 350 gallons per day. While the sample mean is slightly above the claimed value, the difference is not large enough relative to the variability in the data. In part a, the known variance allows

us to use the z-distribution, which assumes more certainty in variability. In part b, the unknown variance adds uncertainty, and the t-distribution compensates with heavier tails. However, the t-test results in an even higher p-value, reinforcing the idea that the sample does not provide sufficient evidence against the claim

CONCLUSION

The analysis supports the public health official's claim that the mean home water usage is 350 gallons per day. Both the z-test (with known variance) and the t-test (with unknown variance) lead to the same conclusion: we fail to reject the null hypothesis. This indicates that the observed sample data is consistent with the claimed average

3

INTRODUCTION

In this study, a nutritionist seeks to determine whether a new diet plan significantly affects body weight. Ten individuals were selected, and their body weight was measured before and after following the diet for one month. A paired t-test is conducted to assess whether there is a statistically significant difference in body weight before and after the diet. This hypothesis test will help us understand whether the diet plan leads to a reduction in body weight

DATA

The data collected includes the body weight (kgs) of 10 participants before and after following the diet plan

METHODOLOGY

Question 3

To evaluate the impact of the diet on body weight, we perform a paired t-test, which compares the mean differences in body weight before and after the diet for each participant. The hypotheses are

H_0 : There is no significant difference in the mean body weight before and after the diet ($\mu_d=0$)

H_1 : There is a significant difference in the mean body weight before and after the diet ($\mu_d \neq 0$)

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

RESULTS

Mean diff: -2.67

std dev of diff 0.45

t: -18.8745

P-value: 0.0000

Reject H_0 , there is a significant difference in body weight after diet

DISCUSSION

The results of the paired t-test indicate a statistically significant weight loss after the diet. The negative t-statistic suggests that participants, on average, lost weight after following the diet. Given the p-value of 0.0000, we reject the null hypothesis and conclude that the diet has a significant impact on body weight. The standard deviation of the differences is quite small, suggesting that the weight changes are consistent across participants, with little variation from the mean

CONCLUSION

Based on the paired t-test, there is strong statistical evidence to support that the diet plan significantly reduces body weight. The average weight loss was 2.67 kg after one month of following the diet. Therefore, we can conclude that the new diet plan is effective in reducing body weight

4

INTRODUCTION

A pharmaceutical company's quality control team aims to verify if their IV fluid filling machine constantly fills bottles within the advertised variance. The manufacturer claims that the variance in the filling volumes should not exceed 4 mL^2 . To assess the validity of this claim, the team randomly selected 15 bottles and measured their volumes. The team now intends to determine whether the machine's performance adheres to the specified variance using a chi-square test for variance

DATA

The dataset consists of the measured volumes of 15 randomly selected bottles, and we are required to calculate sample variance, perform a chi-square test at a significance level of $\alpha=0.01$, $\alpha = 0.01$ to check whether the machine violates the variance specification, and investigate how removing potential outliers affects the test result

METHODOLOGY

Question 4

To assess whether the variance in bottle volumes produced by an IV fluid filling machine exceeds the specified threshold of 4 mL^2 , a chi-square test for population variance was conducted using a random sample of 15 bottles, the sample variance s^2 was computed, then chi-square test statistic was calculated. For a two-tailed test at $\alpha=0.01$ with $df=14$, the critical values from the chi-square distribution are calculated. If the calculated chi-square statistic falls outside the interval, we reject the

null hypothesis. volumes less than 495 mL or greater than 505 mL were considered potential outliers and removed. The test was repeated on the filtered data to examine how outliers affect the conclusion

$$s^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n-1}$$

$$\chi^2 = \frac{(n-1) \cdot s^2}{\sigma^2}$$

RESULTS

Sample variance: 11.6667

Chi square statistic: 40.8333

Chi square critical values: 4.0747 to 31.3193

Reject H0, variance exceeds 4 mL²

After removing outliers

Filtered sample variance: 10.2198

Filtered chi square statistic: 33.2143

Reject H0 (after outlier removal), variance exceeds 4 mL²

DISCUSSION

The calculated sample variance from the original dataset was substantially higher than the claimed 4 mL². The resulting chi-square test statistic was 40.83, which exceeds the upper critical value (31.3193). Thus, there is strong statistical evidence to reject the null hypothesis. Even after removing outliers, the filtered sample variance remained high at 10.22 mL², and the corresponding chi-square test statistic was 33.21, still beyond the acceptable upper limit. This indicates that the observed variation is not just due to extreme values but reflects a more consistent deviation from the expected variance

CONCLUSION

At the 1% significance level, there is strong evidence that the IV fluid filling machine's volume variance exceeds the manufacturer's claimed limit of 4 mL². This conclusion holds both before and after outlier

removal, suggesting that the machine's inconsistency is systemic rather than due to a few anomalous measurements
